

LYSANDER SCHMIDT

Autonomous Vehicles Software Engineer

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on request
lysanderschmidt

Redwood City, CA

EXPERIENCE

Cavalier Racing — Indy Autonomous Challenge

Software Engineer

April 2025 – Ongoing

University of Virginia

- Motion planning/vehicle dynamics: my main contribution has been real-time optimization of velocity profiles along precomputed trajectories, with transparency into the speed limiting factor at every position. (Could be slope, lateral traction, braking before an apex, a hillcrest decreasing downforce, etc.)
- Next step is morphing the racelines in realtime based on reacting to those limiting factors.
- Other examples: code to process LiDAR point clouds of racetracks, to extract surface normals from the roadway surface. (This was necessary to race at Laguna Seca, pictured.) Also to serve pitch and roll references in real time on the car from this data, which allows the gravity component to be accurately subtracted from accelerometer readings, providing better inputs to the localization EKF.

F1Tenth Course (AV Perception, Planning, and Control)

Teaching Assistant

Fall 2025

University of Virginia

- F1Tenth is a feeder course for the autonomous racing team, where students work in groups to program an autonomous 1/10 scale car that races around a classroom-sized racetrack.
- I hold office hours, assist with grading, address configuration and vehicle issues.

Right Option AI

Creator

Summer 2025

rightoption.ai

- Fastest way to get answers (with or without contextual screen info) via voice input with text and speech readback, using the **right Option** key to activate.
- Uses oss-120b/GPT 5 for answers, Gemini Flash for web search, Whisper via Groq for STT, and Amazon Polly or ElevenLabs for TTS.
- Dictation-only mode understands technical jargon, and will copy-edit your spoken words (optionally making them more concise), and paste in your current app.

Stanford Radiochemistry

Lab Assistant

Summer 2024

Stanford School of Medicine

- Increased efficiency and saved coworkers' time by automating data reporting processes (ex: converting machine generated reports to CSV; using GPT-4o to parse scanned handwritten documents for data aggregation).
- Managed radiation containment and safety during transport and delivery of radiopharmaceuticals and acted quickly to mitigate containment breaches. DoT Certified.
- Performed flexible roles such as IT support and medical equipment setup and troubleshooting.



EDUCATION

B.S. in Computer Science

University of Virginia,
School of Engineering

Minor in Leadership

McIntire School of Commerce

May 2026

LANGUAGES & TOOLS

Typescript (& Javascript)

C++

C

Python

Java

Swift (for system calls)

ROS

git

HTML

Excel

node:crypto & Web Crypto(graphy)

Relational DBs (via Prisma, Drizzle)

Bun

Node

Tailscale

AWS

Visual Studio Code

Google Cloud

Linux (Debian, Ubuntu, Alpine)

Docker

ACTIVITIES & CLASSES



UVA Entrepreneurship Cup \$1000 Winner

with two business partners, for
an idea to combat social isolation



Projects with Friends

Created mechanism by which
UValentine.com spread to 8000+
users in ~48 hours.

Built software to allow student
organizations to easily run the
game of assassin.



Selected Courses

Natural Language Processing,
Privacy in the Internet Age,
Effectual Entrepreneurship,
Product and Project Management,
Advanced Business Speaking